Question #1 of 120

Question ID: 1214436

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Moe Girard, CFA, works in a large group that decides on recommendations by consensus. Girard does not always agree with the group consensus, but he is confident in the group's analytical ability. To comply with the Code and Standards when the group issues a recommendation with which he disagrees, Girard:

- **A) does not need to take any action.**
- **B)** must request that his name be removed from the group's report.
- **C)** should include his independent opinion as an appendix to the group's report.

Explanation

Standard V(A) Diligence and Reasonable Basis does not require a Member to dissociate from a group recommendation, as long as the opinion has a reasonable and adequate basis.

For Further Reference:

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #2 of 120

Question ID: 1214426

A professional organization *most appropriately* enforces upon its members:

- **A)** legal standards only.
- **B) ethical standards only.**
- **C)** both legal and ethical standards.

Explanation

Professional organizations adopt codes of ethics that govern their members' behavior. Legal standards are enforced by governments or regulatory agencies.

For Further Reference:

(Study Session 1, Module 1.1, LOS 1.b, LOS 1.f)

CFA® Program Curriculum, Volume 1, page 9

CFA® Program Curriculum, Volume 1, page 15

Question #3 of 120

Question ID: 1214441

Thomas Baker recently passed the Level III CFA examination. Baker is reviewing a draft of the firm's marketing material to be distributed after he receives his CFA charter. One passage reads, "Baker is especially proud of the fact that he passed all three Levels of the exam on his first attempts in three consecutive years." Is this statement in compliance with CFA Institute Standards?

A) **Yes, as long as it is a statement of fact.**

B) No, because it implies that Baker has superior ability.

C) No, because Members or Candidates who passed the exams on their first attempts may not differentiate themselves from those who did not.

Explanation

Stating that Baker passed the exams in consecutive years is acceptable, if in fact he did so, according to Standard VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program.

For Further Reference:

(Study Session 1, Module 3.9, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #4 of 120

Question ID: 1214427

Jimmy Deininger, CFA, manages several client portfolios. One of his clients offers him use of a cabin in a vacation spot because the client's investment results under Deininger's management have exceeded the client's goals. Deininger discloses the gift to his employer. With reference to the Standards of Practice, Deininger:

A) **has complied with the Standards and may accept the gift.**

B) is not permitted to accept the gift because he does not have permission from his employer.

C) has appropriately disclosed the gift to his supervisor, but must also disclose it to his other clients.

Explanation

Gifts from a client are distinguished from gifts from entities attempting to influence the portfolio manager's behavior, such as a broker. Deininger has complied with Standard I(B) Independence and Objectivity because he disclosed the gift from the client to his employer. This requirement is in place so that the employer can monitor the situation to guard against any favoritism towards the gift-giving client. The Standards do not require disclosing this gift to other clients. Permission would be required if the client's gift was to be based on future account performance.

For Further Reference:

(Study Session 1, Module 3.1, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #5 of 120

Question ID: 1214437

Carlos Mendez, CFA, is beginning an investment advisory relationship with a new client and plans to formulate an investment policy statement (IPS) for the client. According to the Standard concerning suitability, Mendez is *least likely* to consider the client's:

A) regulatory and legal circumstances.

B) **conflicts of interest.**

C) performance measurement benchmarks.

Explanation

Under Standard III(C) Suitability, the investment advisor should consider the following in writing an investment policy statement (IPS) for each client: (1) client identification (type and nature of clients, existence of separate beneficiaries, and approximate portion of total client assets); (2) investment objectives (return objectives and risk tolerance); (3) investor constraints (liquidity needs, time horizon, tax considerations, legal and regulatory circumstances, unique needs and preferences); and (4) performance measurement benchmarks. Standard VI(A) Disclosure of Conflicts requires that members and candidates disclose all potential areas of conflict to clients, but this disclosure is not part of a client's IPS.

For Further Reference:

(Study Session 1, Module 3.5, 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #6 of 120

Question ID: 1214442

Which of the following statements about the nine major sections of GIPS is *least accurate*? The major section on:

A) calculation methodology addresses how to determine portfolio and composite returns.

B) presentation and reporting encourages firms to include more information than is required by GIPS when appropriate.

C) **derivatives addresses which valuation methods are appropriate for custom instruments and thinly traded contracts.**

Explanation

Derivatives are not addressed in a specific section of GIPS. The nine major sections are fundamentals of compliance, input data, calculation methodology, composite construction, disclosures, presentation and reporting, real estate, private equity, and wrap fee/separately managed account (SMA) portfolios.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.d)

CFA® Program Curriculum, Volume 1, page 239

Question #7 of 120

Question ID: 1214438

Ann Dunbar, a portfolio manager, wishes to buy stock of Knight Enterprises for her personal account and for clients. Knight is a thinly traded stock. Dunbar believes her own purchase is too small to affect the price but the purchase for clients is likely to increase the price. According to the Code and Standards, when may Dunbar buy the stock for her personal account?

A) **After the buy order for her clients is executed.**

B) At the same time she enters the buy order for her clients.

C) She may not buy the same stock that she buys for her clients.

Explanation

Standard VI(B) Priority of Transactions requires that transactions for clients take precedence over a personal transactions of a member or candidate. Members and candidates should not benefit personally from client transactions, as would occur in this case if the manager enters her personal trade at the same time as the trade for clients. The Standard does not prohibit members and candidates from investing in the same securities they recommend for clients.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #8 of 120

Question ID: 1214429

Telling potential investors that a short-term U.S. Treasury fund contains "guaranteed" securities:

A) does not violate any Standard.

B) violates the Standards by misrepresenting the securities in the fund.

C) violates the Standards by failing to consider the suitability of the fund for potential investors.

Explanation

Standard I(C) Misrepresentation does not prohibit members and candidates from making truthful statements that some investments, such as U.S. Treasury securities, are guaranteed in one way or another. Suitability does not become a concern until the potential clients take investment action.

For Further Reference:

(Study Session 1, Module 3.2, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #9 of 120

Question ID: 1214439

Sean Jones places an order with his investment advisor Lisa Johnson, CFA, to buy 1,000 shares of Orkle Incorporated. Johnson's firm makes a market in Orkle and she executes the trade through her own firm. According to the Code and Standards, Johnson should:

A) disclose her firm's market making activities to Jones.

B) contact her firm's compliance department before accepting the order.

C) decline to execute trades in securities for which her firm makes a market.

Explanation

Standard VI(A) Disclosure of Conflicts states that broker-dealer market making activities must be disclosed to clients.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #10 of 120

Question ID: 1214431

A portfolio manager of a city's police pension fund owes his duty of loyalty to the:

A) city's taxpayers.
B) pension trustees.
C) **plan beneficiaries.**

Explanation

When managing a pension plan or trust, the manager owes his duty of loyalty to the ultimate beneficiaries, not the person or entity that hired the manager.

For Further Reference:

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #11 of 120

Question ID: 1214430

Riley and Smith, a broker-dealer, is bringing to market a secondary offering for All Pro Company. One of the reasons All Pro selected the firm to lead the offering is because Riley and Smith has been a market maker for All Pro's stock for the past five years. The firm is in possession of material nonpublic information relevant to All Pro's offering. To be in compliance with the Code and Standards, Riley and Smith:

A) may not serve as underwriter for the same stock in which it acts as a market maker.
B) **should continue to serve as market maker but take only the contra side of unsolicited customer trades.**
C) should abstain from making a market in All Pro stock during the offering period but may resume market making activities after the offering.

Explanation

The firm should continue making a market but should only carry out unsolicited transactions for clients. A complete withdrawal from market-making activities could be a signal to outsiders that a significant transaction is underway.

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #12 of 120

Question ID: 1214432

Matt O'Neill, CFA, is an advisor for Century Investments, a retail financial services firm. Century has a firmwide policy that its advisors recommend the firm's own investment products to clients unless Century does not offer a product suitable for the client's needs. Can O'Neill follow his firm's policy without violating the Code and Standards?

A) **Yes, if O'Neill discloses this policy to his clients.**
B) Yes, if his firm's offerings are competitive with other available products.

C) No, because the policy conflicts with the Standard on loyalty, prudence, and care.

Explanation

Standard III(A) Loyalty, Prudence, and Care states that members and candidates must inform clients of any limitations that affect their advisory relationships. A policy to favor recommending a firm's own products is an example of such a limitation.

For Further Reference:

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #13 of 120

Question ID: 1214428

After working 20 years on Wall Street, Jim Gentry, CFA, decides to open his own investment firm on Turtle Island, located in the Caribbean. Turtle Island has securities laws that are much less stringent than U.S. laws or the CFA Institute Standards of Professional Conduct. Many of his U.S.-based clients have agreed to keep Gentry as their portfolio manager and move their assets to his new firm. After a few months of operations, Gentry has encountered several instances in which Turtle Island regulations relieve him of disclosing information to investors that he had been required to disclose while working in New York. According to the CFA Institute Code and Standards, Gentry must adhere to the:

A) **Code and Standards or U.S. law, whichever is more strict.**

B) laws of Turtle Island, but disclose any discrepancies to U.S.-based clients.

C) Code and Standards because as a charterholder, he need only adhere to the Code and Standards under all circumstances.

Explanation

Standard I(A) Knowledge of the Law states that when applicable law and the Code and Standards have differing requirements, candidates and members must follow the strictest of the law where they reside, the law where they do business, or the Code and Standards.

For Further Reference:

(Study Session 1, Module 3.1, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #14 of 120

Question ID: 1214440

Wayne Sergeant, CFA, is an independent investment advisor who works with individuals. A longtime client asks Sergeant if he can recommend an attorney. Sergeant refers his client to Jim Chapman, a local attorney who is also a friend of Sergeant's. Previously, Chapman had agreed to perform some legal work for Sergeant in exchange for the referral of new clients. Do Sergeant's actions violate CFA Institute Standards of Professional Conduct?

A) No, because the client is under no obligation and is still free to select another attorney.

B) Yes, because Sergeant is making a recommendation that is not independent and objective.

C) **Yes, because Sergeant did not disclose the nature of his arrangement with Chapman to his client.**

Explanation

Standard VI(C) Disclosure of Conflicts requires members to disclose to their clients any compensation or benefit received by, or paid to, others for the recommendation of services. Sergeant's failure to disclose that he receives legal services for his referral of clients to Chapman is in violation of the Standards.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #15 of 120

Question ID: 1214443

Which of the following statements is *most accurate* regarding the GIPS requirement for definition of the firm?

A) The firm must be the distinct business entity held out to clients.

B) If a firm has offices in different geographical locations, the firm definition may include just the primary location where all the investment decisions are made.

C) The firm definition may include the corporation or a subsidiary of the corporation, but the firm cannot be defined as simply a “division” of the corporation.

Explanation

The GIPS-compliant firm definition must be the corporation, subsidiary, or division that holds itself out to the client as a specific business entity. If the firm has different geographic locations, this firm definition should include all the locations.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.b)

CFA® Program Curriculum, Volume 1, page 236

Question #16 of 120

Question ID: 1214434

Ron Brenner, CFA, manages portfolios for individuals. One of his clients, John Perlman, offers Brenner several inducements above those provided by his employer to motivate superior future performance in managing his portfolio. Brenner notifies his manager via e-mail about the terms of this offer, and his employer grants permission. According to the Standard on additional compensation arrangements, Brenner:

A) must notify “all parties involved,” which includes his other clients.

B) has taken all the actions required to accept the arrangement.

C) should decline this arrangement because it could cause partiality in the handling of other client accounts.

Explanation

Brenner's actions comply with the conditions specified in Standard IV(B) Additional Compensation Arrangements. He notified his employer in writing (e-mail is acceptable) of the terms and conditions of additional compensation arrangement and received permission from his employer. Loyalties to other clients may be affected, but it is the employer's duty to determine this. Nothing in the Standard specifies that "all parties involved" includes other clients.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #17 of 120

Question ID: 1214435

Denise Chavez, CFA, is the senior energy analyst for a major brokerage firm. Chavez is also a social and environmental activist, and is opposed to coal-fired power plants. She has been arrested twice for trespassing during organized pickets at some of these power plants. Chavez has recently accepted a volunteer position as Board member of Greensleeves, a foundation that lobbies governments on environmental issues. The position will involve significant volunteer hours, including some travel. Are Chavez's activities consistent with CFA Institute Standards?

A) Chavez violated the Standards by being arrested, but the volunteer Board position is not a violation.

B) The environmental activism is not a violation, but the Standards prohibit Chavez from accepting the Board position.

C) The activism and subsequent arrests are not a violation, but Chavez must disclose the Board position to her employer.

Explanation

Although Chavez was arrested, Standard I(D) Misconduct is not intended to cover acts of "civil disobedience." Standard IV(A) Loyalty, Chavez has a duty of loyalty to her employer. While she will not be compensated for the Greensleeves' Board position, the duties may be time-consuming and should be discussed with her employer in advance.

For Further Reference:

(Study Session 1, Module 3.2, 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #18 of 120

Question ID: 1214433

With respect to a client's confidential information, if a member or candidate believes a client is engaging in illegal activity, the member should *most appropriately*:

A) preserve the client’s confidentiality.

B) report the client to the appropriate governmental authorities.

C) seek advice from his firm’s legal counsel or compliance department.

Explanation

Guidance for Standard III(E) Preservation of Confidentiality states that members or candidates should seek the advice of compliance personnel or legal counsel about the appropriate actions to take if they suspect illegal activity by clients. Members and candidates must comply with applicable laws, which may require or prohibit disclosure of confidential client information in these circumstances.

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #19 of 120

Question ID: 1214451

Questions 19 to 30 relate to Quantitative Methods. (18 minutes)

George Reilly, CFA, manages the Ivy Foundation portfolio. The Ivy Foundation has a minimum acceptable return of 7%. The current risk-free rate is 6%. Reilly assumes that returns are normally distributed and wants to choose the optimal portfolio for the foundation. The *best* approach Reilly should take is to choose the portfolio that:

A) maximizes the Sharpe ratio.

B) **maximizes the safety-first ratio.**

C) minimizes the standard deviation of returns.

Explanation

Because the Ivy Foundation has a minimum acceptable return that is greater than the risk-free rate, the safety-first ratio is a more suitable criterion than the Sharpe ratio for choosing the optimal portfolio. Given a set of available portfolios, the one that maximizes the safety-first ratio will minimize the probability that the return will be less than the minimum acceptable return if we assume returns are normally distributed. This is the optimal portfolio. Minimizing standard deviation of returns could lead to choosing a portfolio with an expected return below Ivy Foundation's minimum acceptable return.

For Further Reference:

(Study Session 3, Module 9.3, LOS 9.m)

CFA® Program Curriculum, Volume 1, page 537

Question #20 of 120

Question ID: 1214450

A continuous uniform distribution is bounded by zero and 20. The probability of an outcome equal to 12 is *closest* to:

A) **0.00.**

B) 0.05.

C) 0.60.

Explanation

Because the distribution is continuous, the probability of any specific outcome is zero.

For Further Reference:

(Study Session 3, Module 9.1, LOS 9.h)

CFA® Program Curriculum, Volume 1, page 528

Question #21 of 120

Question ID: 1214455

The chi-square test *least likely*:

A) uses a distribution with a lower bound of zero.

B) is used to test whether a variance equals a certain value.

C) **can be used to make inferences even if the population is not normally distributed.**

Explanation

The chi-square test is sensitive to violations of its assumptions. If the population from which the sample is drawn is not normally distributed, then inferences based on the chi-square test will be flawed.

For Further Reference:

(Study Session 3, Module 11.3, LOS 11.j)

CFA® Program Curriculum, Volume 1, page 636

Question #22 of 120

Question ID: 1214449

The joint probability distribution for the return of two retail stocks, A-Marts and Shops R Us, is provided below.

Retail Scenario	Probability	Return for A-Marts	Return for Shops R Us
Good	0.35	0.20	0.10
Average	0.50	0.04	0.02
Poor	0.15	–0.20	–0.10

The covariance between returns for A-Marts and Shops R Us is:

A) less than 0.

B) **at least 0, but less than 0.01.**

C) greater than 0.01.

Explanation

To calculate the covariance, you first must calculate the expected returns (means) for each stock:

Expected return for A-Marts: 0.35(0.20) + 0.50(0.04) + 0.15(–0.20) = 0.06

Expected return for Shops R Us: 0.35(0.10) + 0.50(0.02) + 0.15(–0.10) = 0.03

The covariance is the weighted average of the cross-products:

$$\text{Covariance} = 0.35(0.20 - 0.06)(0.10 - 0.03) + 0.50(0.04 - 0.06)(0.02 - 0.03) + 0.15(-0.20 - 0.06)(-0.10 - 0.03)$$

$$\text{Covariance} = 0.35(0.14)(0.07) + 0.50(-0.02)(-0.01) + 0.15(-0.26)(-0.13) = 0.0086$$

For Further Reference:

(Study Session 2, Module 8.2, LOS 8.k)

CFA® Program Curriculum, Volume 1, page 480

Question #23 of 120

Question ID: 1214444

Frank Jones is considering three separate investments. Investment 1 pays a stated annual interest rate of 6.1%, compounded annually. Investment 2 pays a stated annual interest rate of 6.0%, compounded monthly. Investment 3 pays a stated annual interest rate of 5.9%, compounded quarterly. Which investment should Smith choose?

A) Investment 1.
B) **Investment 2.**
C) Investment 3.

Explanation

Because Investment 1 is compounded annually, its effective annual interest rate is equal to the stated annual rate of 6.1%.

Investment 2 has an effective annual interest rate equal to:

$[1 + (0.06 / 12)]^{12} - 1 = 6.17\%$

Investment 3 has an effective annual interest rate equal to:

$[1 + (0.059 / 4)]^{4} - 1 = 6.03\%$

Jones should choose Investment 2 since it has the highest effective annual interest rate.

For Further Reference:

(Study Session 2, Module 6.1, LOS 6.c)

CFA® Program Curriculum, Volume 1, page 321

Question #24 of 120

Question ID: 1214452

When estimating a population mean or constructing a confidence interval based on the central limit theorem:

A) **the midpoint of a confidence interval is a point estimate of the population parameter.**
B) the degree of significance is the probability that the actual value of the parameter lies within the confidence interval.
C) a point estimate with a 95% degree of confidence is more accurate than a point estimate with a 90% degree of confidence.

Explanation

Confidence intervals for a population mean based on a sample are constructed by multiplying the standard error of a point estimate by a reliability factor, and adding this value to, and subtracting it from, the point estimate. Thus, the point estimate is the midpoint of the confidence interval. The probability that the actual value of the parameter is within a confidence interval is the degree of confidence, which equals one minus the degree of significance. Degrees of confidence or significance apply to confidence intervals but not to point estimates.

For Further Reference:

(Study Session 3, Module 10.2, LOS 10.h, 10.j)

CFA® Program Curriculum, Volume 1, page 579

CFA® Program Curriculum, Volume 1, page 581

Question #25 of 120

Question ID: 1214454

An analyst has a sample of 28 observations with a sample mean of 14.2% and a sample variance of 0.00131. For a test about the population mean, she should use:

A) **a *t*-test if the population is normally distributed.**

B) a *z*-test if the population is normally distributed.

C) either a *z*-test or a t-test regardless of the population distribution..

Explanation

With a small sample size, a normally distributed population, and an unknown population variance, a *t*-test is the appropriate test for a hypothesis about the population mean.

For Further Reference:

(Study Session 3, Module 11.2, LOS 11.g)

CFA® Program Curriculum, Volume 1, page 620

Question #26 of 120

Question ID: 1214447

For a skewed distribution that has excess kurtosis, the minimum percentage of the distribution within three standard deviations of the mean is *closest* to:

A) 68%.

B) **89%.**

C) 99%.

Explanation

Chebyshev's inequality holds regardless of the shape of the distribution. For any $k > 1$, the minimum percentage of the distribution within k standard deviations of the mean is $1 - 1/k^2$. Thus, for 3 standard deviations, the percentage is $\geq 1 - 1/3^2 = 1 - 1/9 = 89\%$.

For Further Reference:

(Study Session 2, Module 7.3, LOS 7.h, 7.l)

CFA® Program Curriculum, Volume 1, page 420

CFA® Program Curriculum, Volume 1, page 422

Question #27 of 120

Question ID: 1214445

Kidra Rao ranks and classifies firms into ten groups based on their interest coverage ratios, lowest to highest. Rao's ranking system is *best* described as:

A) a ratio scale.

B) a nominal scale.

C) **an ordinal scale.**

Explanation

Rao uses an ordinal scale. A nominal scale places data in groups but with no meaningful ranking content. An ordinal scale groups data according to a characteristic that can be ordered, such as grouping stocks based on their rates of return. Ratio scales are the strongest scale of measurement. Ratio scale amounts can be meaningfully added, subtracted, multiplied, and divided. Rao's ranking does not rise to this level (e.g., a group 4 firm does not necessarily have twice the interest coverage of a group 2 firm).

For Further Reference:

(Study Session 2, Module 7.1, LOS 7.a)

CFA® Program Curriculum, Volume 1, page 371

Question #28 of 120

Question ID: 1214446

The harmonic mean of 20, 25, and 30 is *closest* to:

- **A) 24.32.**
- **B)** 24.48.
- **C)** 24.66.

Explanation

Harmonic mean

$$\text{Harmonic mean} = \frac{3}{\frac{1}{20} + \frac{1}{25} + \frac{1}{30}} = 24.32$$

For Further Reference:

(Study Session 2, Module 7.2, LOS 7.e)

CFA® Program Curriculum, Volume 1, page 386

Question #29 of 120

Question ID: 1214453

If a one-tailed *z*-test uses a 5% significance level, the test will reject a:

- **A) true null hypothesis 5% of the time.**
- **B)** false null hypothesis 95% of the time.
- **C)** true null hypothesis 95% of the time.

Explanation

The level of significance is the probability of rejecting the null hypothesis when it is true. The probability of rejecting the null when it is false is the power of a test.

For Further Reference:

(Study Session 3, Module 11.1, LOS 11.c)

CFA® Program Curriculum, Volume 1, page 613

Question #30 of 120

Question ID: 1214448

Tony Borden, CFA, is analyzing the earnings of two companies. For each company, Borden estimates a probability that its earnings will exceed the consensus estimate. To estimate the probability that at least one of the companies will exceed its earnings estimate, Borden should use the:

A) total probability rule.

B) **addition rule of probability.**

C) multiplication rule of probability.

Explanation

The addition rule of probability is used to calculate the probability that at least one of two events will occur: P(A or B) = P(A) + P(B) − P(AB). The total probability rule is used to calculate the unconditional probability of an event given conditional probabilities related to the event: $P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + ... + P(A|B_N)P(B_N)$. The multiplication rule of probability is used to calculate the joint probability that two events will occur together: P(AB) = P(A|B) × P(B).

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.e)

CFA® Program Curriculum, Volume 1, page 463

Question #31 of 120

Question ID: 1214459

Questions 31 through 42 relate to Economics. (18 minutes)

Based on the aggregate demand/aggregate supply model:

A) an inflationary or recessionary gap may exist in the long run.

B) **actual real GDP is equal to potential real GDP in the long run.**

C) no upward or downward pressure on the price level is present at short-run equilibrium.

Explanation

In the short run, real GDP can be less than its full-employment level (a recessionary gap that causes downward pressure on prices) or more than its full-employment level (an inflationary gap that causes upward pressure on prices). In long-run macroeconomic equilibrium, actual real GDP is equal to potential real GDP and there is no upward or downward pressure on the price level.

For Further Reference:

(Study Session 4, Module 14.3, LOS 14.i)

CFA® Program Curriculum, Volume 2, page 161

Question #32 of 120

Question ID: 1214457

A manufacturing plant exhibits diseconomies of scale if long-run average cost (LRAC) is:

A) decreasing as output increases, and the plant is at its minimum efficient scale if LRAC is at its lowest level.

B) decreasing as output increases, and the plant is at its minimum efficient scale if LRAC is decreasing over the entire range of output.

C) **increasing as output increases, and the plant is at its minimum efficient scale if LRAC is at its lowest level.**

Explanation

Diseconomies of scale are present when long-run average cost increases as output increases. The minimum efficient scale is the plant size that produces the quantity of output for which LRAC is at a minimum.

For Further Reference:

(Study Session 4, Module 12.2, LOS 12.f)

CFA® Program Curriculum, Volume 2, page 43

Question #33 of 120

Question ID: 1214464

Placing a tariff on imports of a good is *most likely* to decrease:

A) producer surplus for domestic producers of the good.

B) quantity of the good supplied by domestic producers.

C) **quantity of the good demanded in the domestic market.**

Explanation

Placing a tariff on an imported good increases the good's domestic price, which reduces the quantity demanded. However, the quantity supplied by domestic firms increases with the domestic equilibrium price, as does producer surplus for domestic firms.

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.e)

CFA® Program Curriculum, Volume 2, page 352

Question #34 of 120

Question ID: 1214462

The velocity of transactions in an economy has been increasing rapidly for the past seven years. Over the same time period, the economy has experienced minimal growth in real output. According to the equation of exchange, inflation over the last seven years has:

A) **increased more than the growth in the money supply.**

B) been minimal, consistent with the slow growth in real output.

C) increased at a rate similar to the growth rate in the money supply.

Explanation

The equation of exchange is MV = PY. If velocity (V) is increasing faster than real output (Y), inflation (P) would have to be increasing faster than the money supply (M) to keep the equation in balance.

For Further Reference:

(Study Session 5, Modules 16.1, 16.2, LOS 16.c, 16.k)

CFA® Program Curriculum, Volume 2, page 264

CFA® Program Curriculum, Volume 2, page 283

Question #35 of 120

Question ID: 1214466

Consider two currencies, the VKN and the PKR. The PKR is trading at an annual premium of 2.3% relative to the VKN in the forward market. The 1-year risk-free PKR rate is 3.0%. If no arbitrage opportunities are available, the current 1-year risk-free VKN interest rate is *closest* to:

A) 0.7%.

B) 2.3%.

C) **5.3%.**

Explanation

Because the PKR is trading at a forward premium (the forward VKN/PKR exchange rate is greater than the spot VKN/PKR exchange rate), the VKN interest rate must be greater than the PKR interest rate. VKN should have an interest rate higher than that for PKR by the amount of the forward premium, or approximately 3.0% + 2.3% = 5.3%.

For Further Reference:

(Study Session 5, Module 18.2, LOS 18.h)

CFA® Program Curriculum, Volume 2, page 423

Question #36 of 120

Question ID: 1214458

Wilmer Jones owns several restaurants in different cities. His restaurants compete on quality of food and service, price, and marketing. Competitors can enter and exit his markets, and there are usually several competitors in each market. His market structure can *best* be characterized as:

A) perfect competition.

B) **monopolistic competition.**

C) oligopoly.

Explanation

This is an example of monopolistic competition, because this market has low barriers to entry and exit, and features product differentiation.

For Further Reference:

(Study Session 4, Modules 13.1, 13.4, LOS 13.a, 13.h)

CFA® Program Curriculum, Volume 2, page 64

CFA® Program Curriculum, Volume 2, page 103

Question #37 of 120

Question ID: 1214456

When two goods are complements, the cross elasticity of demand is:

A) positive, and for substitutes the cross price elasticity of demand is negative.

B) negative, and for substitutes the cross price elasticity of demand is negative.

C) **negative, and for substitutes the cross price elasticity of demand is positive.**

Explanation

The cross elasticity of demand for goods that are complements is negative because an increase in the price of one would tend to decrease the quantity demanded of the other. The cross elasticity of demand for substitute goods is positive because an increase in the price of one would tend to increase the quantity demanded of the other.

For Further Reference:

(Study Session 4, Module 12.1, LOS 12.a)

CFA® Program Curriculum, Volume 2, page 9

Question #38 of 120

Question ID: 1214465

With regard to the balance of payments, the purchase of rights to natural resources in a country by foreigners would be *most likely* to affect the country's:

A) **capital account.**

B) current account.

C) financial account.

Explanation

Sales and purchases of non-financial assets in a country are accounted for in the capital account.

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.h)

CFA® Program Curriculum, Volume 2, page 367

Question #39 of 120

Question ID: 1214467

The three-month interest rate in the currency MNO is 4% and the three-month interest rate for the currency PQR is 5%. Based only on this information, the three-month forward MNO/PQR exchange rate:

A) **is less than spot MNO/PQR.**

B) is greater than spot MNO/PQR.

C) may be greater than or less than spot MNO/PQR.

Explanation

Based on the no-arbitrage relationship between spot rates, forward rates, and interest rates, if the interest rate for the base currency is greater than the interest rate for the price currency, the forward exchange rate is less than the spot exchange rate.

For Further Reference:

(Study Session 5, Module 18.2, LOS 18.f)

CFA® Program Curriculum, Volume 2, page 423

Question #40 of 120

Question ID: 1214463

A central bank's policy rate is considered expansionary if it is less than:

A) the central bank’s target inflation rate.

B) the long-term growth rate of real economic output.

C) **the sum of the long-term growth rate of real economic output and the target inflation rate.**

Explanation

Monetary policy is said to be expansionary if the central bank's policy rate is less than the neutral interest rate, which is the sum of the long-term trend rate of real economic growth and the central bank's target inflation rate.

For Further Reference:

(Study Session 5, Module 16.2, LOS 16.m)

CFA® Program Curriculum, Volume 2, page 294

Question #41 of 120

Question ID: 1214460

An analyst who expects the economy to experience stagflation should *most appropriately* recommend investing in:

A) bonds.

B) equities.

C) **commodities.**

Explanation

Stagflation is a period of economic contraction with increasing inflation, typically brought on by a sharp decrease in aggregate supply. Investments in equities tend to perform poorly in an economic contraction due to decreasing profitability of companies. Fixed income investments decrease in price when nominal interest rates increase due to increases in inflation. Commodity prices tend to increase with inflation.

For Further Reference:

(Study Session 4, Module 14.3, LOS 14.l)

CFA® Program Curriculum, Volume 2, page 161

Question #42 of 120

Question ID: 1214461

The consumer price index is *best* described as:

A) the inflation rate for a given period of time.

B) an unbiased estimate of changes in the cost of living.

C) **a weighted average cost for a basket of goods and services.**

Explanation

The consumer price index (CPI) is the average cost of a basket of goods and services, weighted to represent the purchases of a typical household, and indexed to a reference base period. The inflation rate is a percentage change in a price index such as the CPI. Inflation as measured by the CPI is believed to overestimate the actual increase in the cost of living because it does not account for structural changes such as new goods, quality improvements, or consumers shifting their purchases to lower-priced goods.

For Further Reference:

(Study Session 4, Module 15.2, LOS 15.f, 15.g)

CFA® Program Curriculum, Volume 2, page 225

CFA® Program Curriculum, Volume 2, page 227

Question #43 of 120

Question ID: 1214476

Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)

To compute cash collections from customers when converting a statement of cash flows from the indirect to the direct method, an analyst begins with:

A) net income and adds back non-cash expenses.

B) **sales, subtracts any increase in accounts receivable, and adds any increase in unearned revenue.**

C) cost of goods sold, subtracts any increase in accounts payable, adds any increase in inventory, and subtracts any inventory write-offs.

Explanation

To compute cash collections from customers, begin with net sales from the income statement, subtract (add) any increase (decrease) in accounts receivable, and add (subtract) any increase (decrease) in unearned revenue.

For Further Reference:

(Study Session 7, Module 23.3, LOS 23.g)

CFA® Program Curriculum, Volume 3, page 211

Question #44 of 120

Question ID: 1214472

Items that appear in other comprehensive income, but are excluded from the income statement, include:

A) losses due to expropriation of assets.

B) **gains and losses due to foreign currency translation.**

C) unrealized gains and losses on trading securities.

Explanation

Other comprehensive income includes unrealized gains and losses on available-for-sale securities, foreign currency translation gains and losses, minimum pension liability adjustments, and unrealized gains and losses on derivatives used for cash flow hedging.

Unrealized gains and losses on held-for-trading securities are included in net income on the income statement. Losses due to expropriation of assets would be included in net income, most likely as an unusual or infrequent item.

For Further Reference:

(Study Session 7, Module 21.5, LOS 21.l)

CFA® Program Curriculum, Volume 3, page 113

Question #45 of 120

Question ID: 1214485

Other things equal, which of the following conditions would place a company highest on a spectrum of financial reporting quality?

- **A)** **Reported earnings that are not sustainable.**
- **B)** Efforts by management to keep net income steady over time.
- **C)** Financial statements that reflect the company's economic activities accurately but are not in compliance with accounting principles.

Explanation

Earnings quality may be low in a period because of one-time gains that do not otherwise call a company's financial reporting quality into question. Earnings smoothing or reporting that does not comply with generally accepted accounting principles represents a lower quality of financial reporting.

For Further Reference:

(Study Session 9, Module 29.1, LOS 29.b)

CFA® Program Curriculum, Volume 3, page 541

Question #46 of 120

Question ID: 1214480

A company purchases an asset in the first quarter and decides to capitalize the asset. Compared to expensing the asset cost, capitalizing the asset cost will result in higher cash flows in the first quarter from:

- **A)** investing.
- **B)** financing.
- **C)** **operations.**

Explanation

Capitalizing the cost of the asset results in higher CFO and lower CFI in the period of the purchase, compared to expensing the entire cost. If the cost is expensed, the cash outflow is classified as CFO, but if the asset is capitalized, the cash outflow is classified as CFI. Cash flow from financing is not affected by the decision to capitalize.

For Further Reference:

(Study Session 8, Module 26.1, LOS 26.c)

CFA® Program Curriculum, Volume 3, page 386

Question #47 of 120

Question ID: 1214470

On a firm's income statement, sales minus cost of goods sold, minus selling, general, and administrative expenses, is *most appropriately* referred to as:

A) gross profit.
B) **operating profit.**
C) income before tax.

Explanation

This difference describes operating profit.

For Further Reference:

(Study Session 7, Module 21.1, LOS 21.a)

CFA® Program Curriculum, Volume 3, page 72

Question #48 of 120

Question ID: 1214483

XYZ Company has decided to issue $10 million of unsecured bonds. If issued today, the 4% semi-annual coupon bonds would require a market interest rate of 12%. Under U.S. GAAP, how will these bonds affect XYZ's statement of cash flows?

A) **The coupon payments will decrease operating cash flow each year and the discount will decrease financing cash flow at maturity.**
B) The periodic interest expense will decrease operating cash flow and the discount will decrease financing cash flow at maturity.
C) The coupon payments and the discount amortization will decrease financing cash flow each year.

Explanation

It is the coupon payment, not the interest expense, that results in an outflow of cash. The difference between the coupon payment and interest expense is the discount amortization. The amortization does not result in a cash outflow. Under U.S. GAAP, the coupon payment is reported as an operating cash flow. The discount, when paid at maturity, is reported as a financing cash flow.

For Further Reference:

(Study Session 8, Modules 28.1, 28.2, LOS 28.a, 28.b)

CFA® Program Curriculum, Volume 3, page 490

CFA® Program Curriculum, Volume 3, page 494

Question #49 of 120

Question ID: 1214479

Data for a manufacturing industry indicate that inventories of work in progress are increasing faster than sales. This is *most likely* to indicate that:

A) the business cycle is at a peak.
B) inventory is becoming obsolete.
C) **firms expect demand to increase.**

Explanation

An increase in work-in-progress inventory relative to sales is likely to result from firms increasing production because they expect an increase in demand. An increase in finished goods inventories relative to sales would be more likely to indicate a decrease in demand that may be caused by obsolete inventory or a business cycle peak.

For Further Reference:

(Study Session 4, Module 15.1, LOS 15.b and Study Session 8, Module 25.4, LOS 25.j)

CFA® Program Curriculum, Volume 2, page 200

CFA® Program Curriculum, Volume 3, page 339

Question #50 of 120

Question ID: 1214473

A classified balance sheet categorizes assets and liabilities based on whether they are:

A) **current or non-current items.**
B) measured at cost or fair value.
C) internally generated or acquired.

Explanation

Classified balance sheets have categories for current assets, non-current assets, current liabilities, and non-current liabilities.

For Further Reference:

(Study Session 7, Module 22.2, LOS 22.c)

CFA® Program Curriculum, Volume 3, page 129

Question #51 of 120

Question ID: 1214484

Under IFRS, a lessor will remove the leased asset from its balance sheet and record interest income from the lease only if the lease is classified as:

A) **a finance lease.**
B) a sales-type lease.
C) an operating lease.

Explanation

Under IFRS, a lessor will classify a lease as either an operating lease or a finance lease. If it is classified as a finance lease, the leased asset is removed from the lessor's balance sheet and interest income is recognized over the life of the lease.

For Further Reference:

(Study Session 8, Module 28.4, LOS 28.h)

CFA® Program Curriculum, Volume 3, page 511

Question #52 of 120

Question ID: 1214474

Jansen Co., a manufacturer of high-end sports equipment, earned $45 million in net income for the year. The company paid out $1.30 per share in dividends. Jansen issued 500,000 shares at the beginning of the year at $20 (1 million shares were outstanding before the issuance). The market value of Jansen's trading securities decreased by $2.4 million. The increase in Jansen's stockholders' equity is *closest* to:

A) $43 million.

B) $51 million.

C) **$53 million.**

Explanation

The unrealized loss on trading securities is reflected in net income. The total change in stockholder's equity is:

$45,000,000 − [(1,000,000 + 500,000 shares) × $1.3/share] + (500,000 × $20/share) = $53,050,000

For Further Reference:

(Study Session 7, Module 22.7, LOS 22.f)

CFA® Program Curriculum, Volume 3, page 157

Question #53 of 120

Question ID: 1214468

Mullins Company's financial statements include an auditor's report with a qualified opinion. This *most likely* implies that the:

A) auditor is reasonably assured that the financial statements are free of material errors.

B) **financial statements include exceptions to the applicable accounting standards but are presented fairly.**

C) financial statements are materially out of compliance with the applicable accounting standards and are not presented fairly.

Explanation

An auditor will issue a qualified opinion if the financial statements include exceptions to applicable accounting standards and will explain the nature and effect of these exceptions. An auditor will issue an adverse opinion if the financial statements are not presented fairly.

For Further Reference:

(Study Session 6, Module 19.2, LOS 19.d)

CFA® Program Curriculum, Volume 3, page 28

Question #54 of 120

Question ID: 1214475

Use of the indirect method of presenting cash flows from operating activities:

A) is encouraged by both the IASB and FASB.

B) **illustrates the reasons for the difference between net income and operating cash flow.**

C) requires disclosure of the cash flows that would be presented using the direct method.

Explanation

An argument in favor of using the indirect method is that it links the income statement with the cash flow statement by focusing on the differences between net income and operating cash flow. Standard setting bodies permit either method but encourage use of the direct method. Companies that use the direct method must provide a disclosure that reconciles net income with cash flow from operations, similar information to what would be presented under the indirect method.

For Further Reference:

(Study Session 7, Module 23.1, LOS 23.d)

CFA® Program Curriculum, Volume 3, page 188

Question #55 of 120

Question ID: 1214471

A firm has undertaken a contract with an estimated total cost of $500 million at a price of $800 million. At the end of the first reporting period, the firm has devoted resources of $180 million to the project. The customer has been billed for $250 million and made payments of $160 million. The amount of revenue the firm should record for the period is *closest* to:

A) $180 million.
B) $250 million.
C) **$290 million.**

Explanation

Using the percentage of total costs incurred to date as an estimate of the portion of the performance obligations completed, revenue should be (180 / 500) × $800 million = $288 million.

For Further Reference:

(Study Session 7, Module 21.2, LOS 21.c)

CFA® Program Curriculum, Volume 3, page 79

Question #56 of 120

Question ID: 1214478

Inventory cost is *most likely* to include:

A) storage costs for finished goods until they are actually sold.
B) shipping cost for delivery to the customer.
C) **an allocation of fixed production overhead.**

Explanation

An allocation of fixed production overhead based on normal production capacity is included in inventory cost. Neither storage costs that are not required as part of the production process nor shipping costs for delivery to the customer are included in inventory cost.

For Further Reference:

(Study Session 8, Module 25.1, LOS 25.a)

CFA® Program Curriculum, Volume 3, page 313

Question #57 of 120

Question ID: 1214469

Which of the following statements *most accurately* describes the general features of financial statements under IFRS?

A) All of the required financial statements are prepared using accrual accounting.

B) **Assets may not be offset against liabilities unless specifically permitted or required by a standard.**

C) Prior-period information may only be presented when specifically permitted or required by a standard.

Explanation

One of the general requirements stated in IAS No. 1 is that firms not offset assets against liabilities unless a specific standard permits or requires it. The statement of cash flows is not prepared using accrual accounting. IAS No. 1 states that firms should present comparative information for prior periods unless a specific standard states otherwise.

For Further Reference:

(Study Session 6, Module 20.2, LOS 20.d)

CFA® Program Curriculum, Volume 3, page 58

Question #58 of 120

Question ID: 1214482

Dot Corporation uses accelerated depreciation for tax purposes and straight-line depreciation for financial reporting. The company has a large cash position which is invested in tax-free municipal bonds. With regard to Dot's financial statements and tax reporting:

A) both the interest income and the depreciation method will necessitate the use of a valuation allowance account.

B) the interest income will result in a deferred tax asset and the depreciation method will result in a deferred tax liability.

C) **the depreciation expense causes a temporary difference between income tax expense and taxes payable, and the interest income creates a permanent difference.**

Explanation

The interest income from municipal bonds is a permanent difference; thus, no deferred taxes are created and the difference is reflected in the company's effective tax rate. The different depreciation methods result in temporary differences that are expected to reverse. In the case of depreciation, a deferred tax liability is created. Valuation allowance accounts only apply to deferred tax assets and are created when it becomes probable that the company will not have enough future income to realize the full value of the deferred tax assets.

For Further Reference:

(Study Session 8, Module 27.5, LOS 27.f)

CFA® Program Curriculum, Volume 3, page 462

Question #59 of 120

Question ID: 1214477

Bivac Corp. has been experiencing a declining return on equity over the past few years. Selected financial statement ratios for Bivac appear below:

	Prior Year	Current Year
Tax Burden	0.60	0.62
Interest Burden	0.80	0.81
EBIT Margin	0.26	0.26
Asset Turnover	1.06	1.06
ROE	0.15	0.14

What is the *most likely* reason for the decline in Bivac's ROE?

A) Leverage has declined.
B) The tax rate has increased.
C) Net profit margin has declined.

Explanation

ROE can be broken out as:

ROE = Tax burden × Interest burden × EBIT Margin × Asset Turnover × Leverage

Prior Year: 0.15 = 0.60 × 0.80 × 0.26 × 1.06 × Leverage

Current Year: 0.14 = 0.62 × 0.81 × 0.26 × 1.06 × Leverage

Solving the equation for leverage reveals that the measure has decreased from 1.13 in the prior year to 1.01 in the current year. This indicates Bivac is using less debt in its capital structure and is the most likely reason the company's ROE has declined.

The company's net profit margin has increased:

Net profit margin (Prior Year): 0.60 × 0.80 × 0.26 = 0.12

Net profit margin (Current Year): 0.62 × 0.81 × 0.26 = 0.13

The company's tax rate has decreased from 0.40 = (1 − 0.60) to 0.38 = (1 − 0.62).

For Further Reference:

(Study Session 7, Module 24.4, LOS 24.d)

CFA® Program Curriculum, Volume 3, page 279

Question #60 of 120

Question ID: 1214481

Stone Development Company owns four office buildings and a tract of raw land. Stone occupies one of the buildings, collects rental income from the other three buildings, and is holding the land for capital appreciation. Under IFRS, which of these assets should Stone classify as investment property on its balance sheet?

A) All of these assets.
B) Only the land held for capital appreciation.
C) The land and the buildings that generate rental income.

Explanation

Investment property is defined under IFRS as property held for the purpose of earning rental income, capital appreciation, or both. Owner-occupied property is not classified as investment property.

For Further Reference:

(Study Session 8, Module 26.4, LOS 26.n)

CFA® Program Curriculum, Volume 3, page 428

Question #61 of 120

Question ID: 1214489

Questions 61 through 72 relate to Corporate Finance. (18 minutes)

ChemCo is evaluating a project based on the principles of capital budgeting. ChemCo should accept the project if it has a:

A) net present value equal to zero.

B) **profitability index greater than one.**

C) required rate of return greater than its internal rate of return.

Explanation

Several decision criteria are available when evaluating a standalone project. These criteria include net present value (NPV), profitability index (PI), and internal rate of return (IRR). The decision rules for these measures are as follows.

NPV:	If NPV > 0, accept project
	If NPV < 0, reject project
PI:	If PI > 1, accept project
	If PI < 1, reject project
IRR:	If IRR > required return, accept project
	If IRR < required return, reject project

For Further Reference:

(Study Session 10, Modules 32.1, 32.2, LOS 34.d)

CFA® Program Curriculum, Volume 4, page 52

Question #62 of 120

Question ID: 1214496

Mentemeyer Corporation is a small firm that needs to increase short-term liquidity but has weak credit. The source of short-term financing that would *most likely* be available to Mentemeyer is:

A) commercial paper.

B) **nonbank finance companies.**

C) a revolving credit agreement.

Explanation

Nonbank finance companies are a source of short-term financing for smaller firms and firms with lower credit ratings. Commercial paper issuance and revolving credit agreements are typically only available to larger corporations with high credit ratings.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.g)

CFA® Program Curriculum, Volume 4, page 189

Question #63 of 120

Question ID: 1214488

Johnson's Jar Lids is deciding whether to begin producing jars. Johnson's pays a consultant $50,000 for market research that concludes Johnson's sales of jar lids will increase by 5% if it also produces jars. In choosing the cash flows to include when evaluating a project to begin producing jars, Johnson's should:

A) include both the cost of the market research and the effect on the sales of jar lids.
B) include the cost of the market research and exclude the effect on the sales of jar lids.
C) exclude the cost of the market research and include the effect on the sales of jar lids.

Explanation

Sunk costs should be excluded from cash flows, as they are costs that cannot be avoided even if the project is not undertaken. Externalities, such as positive or negative effects of accepting a project on sales of the company's existing products, should be included in the cash flows.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.b)

CFA® Program Curriculum, Volume 4, page 50

Question #64 of 120

Question ID: 1214497

While analyzing HMS Inc., Fred Browne notes that the company's liquidity as measured by its quick ratio has decreased over time while its current liabilities have remained constant. This could be explained by:

A) a decrease in inventory.
B) an increase in marketable securities.
C) a decrease in accounts receivable.

Explanation

The quick ratio is defined as: (cash + marketable securities + accounts receivable) / current liabilities. If current liabilities have remained constant, cash, marketable securities, or accounts receivable must have decreased. Inventory is not included in the quick ratio.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.b)

CFA® Program Curriculum, Volume 4, page 159

Question #65 of 120

Question ID: 1214490

Over the next year, Thatherton Co. is expecting their marginal tax rate to increase by 5%. Also, over the next 12 months, Thatherton plans to undertake several expansion projects significantly more risky than previous projects. Thatherton Co.'s current capital structure includes 40% debt and 60% equity. Which of the following statements *correctly* summarizes the effect these changes will have on the company's marginal cost of capital?

A) The increasing tax rate will increase the MCC.

B) **The riskier projects will increase the MCC.**

C) Both the increasing tax rate and the riskier projects will increase the MCC.

Explanation

Recall that the marginal cost of capital (MCC) is the cost of the last dollar of new capital raised by the firm. Marginal cost increases as increasing amounts of capital are raised during a set period. In general, firms in riskier businesses, or with riskier projects, have higher costs of common equity and thus higher MCC and WACC. The increase in the tax rate would reduce the after-tax cost of debt, reducing the MCC.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.a, 33.b)

CFA® Program Curriculum, Volume 4, page 78

CFA® Program Curriculum, Volume 4, page 78

Question #66 of 120

Question ID: 1214486

Which of the following is *most likely* to create an incentive for senior managers to act in the best interests of shareholders?

A) A one-tier board structure.

B) **The threat of a hostile takeover.**

C) A cash bonus for meeting short-term performance targets.

Explanation

The threat of a hostile takeover, one in which senior managers are replaced, can act as a strong incentive for senior managers to act in shareholders' interests by growing the value of the company's equity shares. Companies where management has followed their own interests rather than those of shareholders can show slow growth and low return on assets and attract an outside buyer who hopes to profit from improving company operations and strategy.

For Further Reference:

(Study Session 10, Module 31.2, LOS 31.g, 31.i)

CFA® Program Curriculum, Volume 4, page 25

CFA® Program Curriculum, Volume 4, page 31

Question #67 of 120

Question ID: 1214494

Operating risk is *most likely* to increase as a result of:

A) an increase in sales risk.

B) increased variability of costs.

C) an increase in fixed production costs.

Explanation

Operating risk refers to uncertainty about operating earnings arising from fixed production (operating) costs.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.a)

CFA® Program Curriculum, Volume 4, page 126

Question #68 of 120

Question ID: 1214491

The following data are reported for Moving Vans, Inc.:

Dividend yield	5%
Dividend payout	20%
Return on equity	15%

Assuming Moving Vans' dividend yield, dividend payout, and return on equity will remain constant indefinitely, the cost of equity capital is *closest* to:

A) 15%.

B) 17%.

C) 19%.

Explanation

Using the constant growth dividend model, the price for a common stock is:

$$P_0 = \frac{D_1}{k_{ce} - g}$$

Solving for k_{ce}:

$$k_{ce} = \frac{D_1}{P_0} + g$$

where:

P_0 = current stock price

D_1 = year end expected dividend

k_{ce} = cost of common equity capital

g = sustainable (constant) growth of equity, earnings, and dividends

The sustainable growth for a company's dividends equal the ROE times the earnings retention rate. The earnings retention rate equals 1 minus the dividend payout rate. Therefore:

g = 0.15 × (1 − 0.20) = 0.12

$$k_{ce} = \frac{D_1}{P_0} + g = 0.05 + 0.12 = 0.17 = 17\%$$

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.h)

CFA® Program Curriculum, Volume 4, page 88

Question #69 of 120

Question ID: 1214493

The asset beta of a firm equals its equity beta if:

- **A) the company has no debt.**
- **B)** the company has no equity.
- **C)** the company's debt equals its equity.

Explanation

The formula for the asset beta is:

$$\beta_{\text{asset}} = \beta_{\text{equity}} \left[\frac{1}{1 + \left(\frac{D(1-t)}{E} \right)} \right]$$

Therefore, the two betas are identical only if the company has no debt in its capital structure (D = 0). If the company has no debt, then the asset beta must equal the equity beta.

For Further Reference:

(Study Session 10, Module 33.2, LOS 33.i)

CFA® Program Curriculum, Volume 4, page 94

Question #70 of 120

Question ID: 1214487

Joanne Soto is responsible for implementing her firm's environmental, social, and governance (ESG) investing goals. Soto decides on a best-in-class approach to ESG investing. Soto will *most likely* instruct her firm's portfolio managers to use:

- **A) positive screening.**
- **B)** full integration.
- **C)** active ownership.

Explanation

The best-in-class approach to ESG implementation is an example of positive screening. Full integration refers to including ESG factors in fundamental analysis. Active ownership is using share voting and access to company management to pursue ESG objectives.

For Further Reference:

(Study Session 10, Module 31.2, LOS 31.k)

CFA® Program Curriculum, Volume 4, page 40

Question #71 of 120

Question ID: 1214495

Archer Company produces components that sell for $3.50 per unit. Archer has fixed financing costs of $125,000 and fixed operating costs of $275,000. Archer's variable costs are $2.50 per unit. Archer will:

A) break even if it sells 275,000 units.

B) **experience a net loss if it sells 325,000 units.**

C) earn positive net income if it sells 375,000 units.

Explanation

Archer's breakeven quantity of sales is $\frac{\$275{,}000+\$125{,}000}{\$3.50-\$2.50}$ = 400,000 units. Archer will experience a net loss at any quantity of sales less than 400,000.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.d)

CFA® Program Curriculum, Volume 4, page 143

Question #72 of 120

Question ID: 1214492

A firm has one actively traded bond issue outstanding, with a 6% coupon and a yield to maturity of 5%. When estimating the firm's weighted average cost of capital (WACC), the appropriate after-tax cost of debt capital should *most likely* be:

A) between 5% and 6%.

B) **less than 5%.**

C) equal to 6%.

Explanation

Yield to maturity is an appropriate estimate of a firm's before-tax cost of capital. Its after-tax cost of capital may be estimated as YTM × (1 – tax rate) and will be less than the before-tax cost of capital, as long as the firm faces a positive tax rate, which is the most likely case.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.d)

CFA® Program Curriculum, Volume 4, page 82

Question #73 of 120

Question ID: 1214500

Questions 73 through 85 relate to Equity Investments. (19.5 minutes)

Which type of stock index must be adjusted for stock splits?

A) Equal weighted index.

B) **Price weighted index.**

C) Market capitalization weighted index.

Explanation

When computing any price-weighted index, the denominator must be adjusted to take stock splits into account.

For Further Reference:

(Study Session 12, Module 37.1, LOS 37.d)

CFA® Program Curriculum, Volume 4, page 285

Question #74 of 120

Question ID: 1214502

Aros Funds manages a family of mutual funds and employs a team of fundamental analysts, who research firms by analyzing financial statements and SEC filings. Under which form(s) of the efficient market hypothesis (EMH) would Aros Funds have the potential to achieve positive risk-adjusted returns consistently using fundamental analysis?

A) **Weak form only.**
B) Semistrong form and weak form.
C) No form of the EMH is consistent with earning positive risk-adjusted returns using fundamental analysis.

Explanation

Fundamental analysis (i.e., analysis of public non-market data) can produce positive risk-adjusted returns if markets are weak-form efficient but not semistrong-form efficient. Under the semistrong form of the EMH, fundamental analysis cannot consistently achieve positive risk-adjusted returns.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.e)

CFA® Program Curriculum, Volume 4, page 330

Question #75 of 120

Question ID: 1214504

An investor in a sponsored depository receipt (DR):

A) **holds the voting rights for the DR shares.**
B) must obtain the foreign currency in which the DR is traded.
C) should be familiar with market procedures and regulations in the DR issuer's country.

Explanation

In a sponsored DR, voting rights of the shares are held by the investor. In an unsponsored DR, voting rights are retained by the depository bank. An advantage of DR shares over direct investments in foreign companies is that DR shares trade in the investor's domestic market and currency.

For Further Reference:

(Study Session 13, Module 39.2, LOS 39.d)

CFA® Program Curriculum, Volume 4, page 365

Question #76 of 120

Question ID: 1214508

Ian Goode, CFA, is analyzing the price of the preferred stock of MegaGym. Goode estimates that MegaGym's earnings growth rate over the next five years will be 20%, and that MegaGym's earnings will then grow at a sustainable rate of 5%. The *most appropriate* method for Goode to value MegaGym's preferred stock is to:

A) use a justified price-to-earnings multiple.

B) use a multistage dividend discount model with 20% growth for five years and 5% thereafter.

C) **divide the preferred dividend by the required rate of return on MegaGym's preferred stock.**

Explanation

The value of preferred stock is the preferred dividend divided by the required rate of return on the preferred. Earnings growth rates do not factor into the valuation of preferred stock since the dividend is typically fixed. Therefore, neither a historical price-to-earnings model nor a multistage dividend discount model is appropriate.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.f)

CFA® Program Curriculum, Volume 4, page 456

Question #77 of 120

Question ID: 1214505

Industry rotation is *best* described as the:

A) **adjusting the industry weights in a portfolio based on the current stage of the business cycle.**

B) recommended practice of periodically changing the industries that investment analysts are assigned to cover.

C) long-term trend of talented managers and employees exiting mature and declining industries and entering embryonic and growth industries.

Explanation

Industry rotation is an active management strategy of overweighting or underweighting industries, compared to their strategic allocation weights, based on the stage of the business cycle.

For Further Reference:

(Study Session 13, Module 40.1, LOS 40.a)

CFA® Program Curriculum, Volume 4, page 388

Question #78 of 120

Question ID: 1214503

Sacco, Inc., has nine directors on its board. Three seats are elected annually using a cumulative voting system. An investor who owns 600 shares of Sacco common stock may give a maximum of:

A) 600 votes to a single board candidate.

B) **1,800 votes to a single board candidate.**

C) 600 votes to a candidate for each board seat.

Explanation

Under cumulative voting, with three board seats up for election, the shareholder can cast up to 1,800 votes for one candidate. In a statutory voting system, an investor who holds 600 shares can cast 600 votes in the election for each seat.

For Further Reference:

(Study Session 13, Module 39.1, LOS 39.b)

CFA® Program Curriculum, Volume 4, page 358

Question #79 of 120

Question ID: 1214501

Which of the following types of index is *least likely* to require frequent reconstitution of constituent securities?

A) **Equity index.**

B) Commodity index.

C) Fixed income index.

Explanation

Equity indexes typically require reconstitution only in response to corporate events, such as mergers or bankruptcies. Commodity indexes, which use futures contracts as their constituent securities, and fixed income indexes require frequent reconstitution as futures contracts expire and bonds mature.

For Further Reference:

(Study Session 12, Module 37.2, LOS 37.f, 37.i, 37.j)

CFA® Program Curriculum, Volume 4, page 293

CFA® Program Curriculum, Volume 4, page 300

CFA® Program Curriculum, Volume 4, page 303

Question #80 of 120

Question ID: 1214498

An electronic crossing network is *best* described as:

A) a price-driven market.

B) a quote-driven market.

C) **an order-driven market.**

Explanation

A crossing network is an example of an order-driven market. Orders are batched together and crossed (matched) at specific times during the trading day at prices based on those of another exchange. Price-driven markets and quote-driven markets are other terms for dealer or over-the-counter markets.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.j)

CFA® Program Curriculum, Volume 4, page 256

Question #81 of 120

Question ID: 1214507

An investor places a market order to buy a stock on the holder-of- record date for the stock's next dividend. Is the investor entitled to receive this dividend?

- **A)** No, because the order is placed after the declaration date.
- **B) No, because the order is settled after the holder-of-record date.**
- **C)** Yes, because the order is executed on the holder-of-record date.

Explanation

To receive the next dividend, an investor must buy a stock before its ex-dividend date, which is one or two business days before the holder-of-record date. This one- or two-day period allows orders executed before the ex-dividend date to be settled by the holder-of-record date. The buyer would be entitled to the dividend if the order was executed after the declaration date but before the ex-dividend date.

For Further Reference:

(Study Session 13, Module 41.1, LOS 41.c)

CFA® Program Curriculum, Volume 4, page 450

Question #82 of 120

Question ID: 1214499

Kate Johnson, CFA, owns shares of a stock that currently trades at $15. If Johnson wants to buy more shares if the price increases to $17, she should enter a:

- **A) stop buy order at $17.**
- **B)** limit order to buy at $17.
- **C)** market order to buy at $17.

Explanation

An order to buy if a price increases to a specified level is a stop buy order. A limit order at $17 will execute immediately if the market price is $15. A market order does not specify a price, but is executed at the prevailing market price.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.g, 36.h)

CFA® Program Curriculum, Volume 4, page 246

CFA® Program Curriculum, Volume 4, page 246

Question #83 of 120

Question ID: 1214506

In the industry life cycle model, the threat of new entry into an industry is greatest during the:

A) mature stage.

B) **growth stage.**

C) embryonic stage.

Explanation

New competitors are most likely to enter an industry during the growth stage of the industry life cycle. New entrants are less of a threat in the embryonic stage, when growth is slow and customer acceptance of the new product or service is highly uncertain. In the mature stage, the industry tends toward an oligopoly as competitors consolidate.

For Further Reference:

(Study Session 13, Module 40.2, LOS 40.h)

CFA® Program Curriculum, Volume 4, page 414

Question #84 of 120

Question ID: 1214510

An analyst develops the following information to value a common stock.

- Last year's earnings per share = \$4.00
- Real risk-free rate = 4%
- Inflation premium = 5%
- Return on equity (ROE), expected to remain constant in the future = 10%
- Dividend payout, expected to remain stable in the future = 30%
- Stock's beta = 1.4
- Expected market return = 14%

The value per share is *closest* to:

A) **\$14.39.**

B) \$21.28.

C) \$31.39.

Explanation

$RFR_{nominal} = (1 + RFR_{real})(1 + IP) - 1 = (1.04)(1.05) - 1 = 1.0920 - 1 = 0.0920 = 9.20\%$

Using the CAPM, the required rate of return (k_e) = $RFR_{nominal} + \beta(R_{mkt} - RFR_{nominal}) = 9.20\% + 1.4(14.0\% - 9.2\%) = 9.20\% + 6.72\% = 15.92\%$

The retention ratio (RR) = 1 − dividend payout ratio = 1 − 0.30 = 0.70

The growth rate (g_c) = (RR)(ROE) = (0.70)(10%) = 7.00%

$D_0 = E_0$(dividend payout) = \$4.00(0.30) = \$1.20

Next year's dividend (D_1) = $D_0(1 + g_c)$ = \$1.20(1 + 0.07) = 1.284

$P_0 = D_1 / (k_e - g) = 1.284 / (0.1592 - 0.07) = 14.39$

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.j)

SchweserNotes, Book 4 page 215

CFA® Program Curriculum, Volume 4, page 468

Question #85 of 120

Question ID: 1214509

At the end of the last 12-month period, Romano's Italian Foods had net income of $16.68 million and equity of $115 million. Romano's declared a $7.5 million dividend for the year. Using internally generated funds, Romano's can grow its equity by approximately:

A) 8.0% per year.
B) 10.0% per year.
C) 14.5% per year.

Explanation

g = ROE × retention rate = [16.68 / 115] × [1 − (7.5 / 16.68)] = 0.145 × (1 − 0.45) = 7.975%. This growth rate represents the rate at which a company can grow its equity using internally generated funds.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #86 of 120

Question ID: 1214514

Questions 86 through 99 relate to Fixed Income. (21 minutes)

A bond pays a quarterly coupon of 8% minus one-half of annual 90-day LIBOR. This bond is *most accurately* classified as a:

A) leveraged instrument.
B) participation instrument.
C) yield enhancement instrument.

Explanation

An inverse floater is classified as a leveraged instrument. The example given here is a deleveraged inverse floater because the multiplier for the reference rate is less than one.

For Further Reference:

(Study Session 14, Module 43.2, LOS 43.h)

CFA® Program Curriculum: Volume 5, page 87

Question #87 of 120

Question ID: 1214518

Which of the following mortgage-backed securities is *most likely* to feature credit tranching?

A) Collateralized mortgage obligations.
B) Commercial mortgage-backed securities.
C) Agency residential mortgage-backed securities.

Explanation

Commercial mortgage-backed securities often feature credit tranching in which subordinated tranches are the first to absorb credit losses. Sequential-pay CMOs employ time tranching in which all principal payments flow to Tranche 1 up to its principal amount, then to Tranche 2 up to its principal amount, and so on. Agency RMBS are pass-through securities and do not feature credit tranching or time tranching.

For Further Reference:

(Study Session 14, Module 45.1, LOS 45.c)

CFA® Program Curriculum: Volume 5, page 185

Question #88 of 120

Question ID: 1214511

A waterfall structure in a securitized bond issue:

A) is a form of external credit enhancement.

B) allows the entire issue to obtain a better credit rating.

C) **gives some bondholders a higher priority of claims than others.**

Explanation

In a securitized bond issue with a waterfall structure, the bonds are issued in tranches with varying levels of seniority. Any losses arising from the underlying assets are absorbed first by the tranches with the lowest seniority. Thus, tranches have different levels of default risk and therefore are likely to have different credit ratings. The structure is a form of internal credit enhancement.

For Further Reference:

(Study Session 14, Module 42.1, LOS 42.d)

CFA® Program Curriculum: Volume 5, page 21

Question #89 of 120

Question ID: 1214515

Bond X and Bond Y were issued at a premium to par value three years ago. Bond X matures in five years, and Bond Y matures in ten years. Both bonds carry the same credit rating. Bond X has a coupon of 7.25%, and Bond Y has a coupon of 8.00%. If the yield to maturity for both bonds is 7.60% today:

A) both bonds are priced at a premium.

B) Bond X is priced at a premium, and Bond Y is priced at a discount.

C) **Bond X is priced at a discount, and Bond Y is priced at a premium.**

Explanation

Because Bond X has a coupon rate that is below the required yield, it will trade at a discount to par. Bond Y, with a coupon rate greater than the required yield, will trade at a premium to par. The fact that both bonds were issued at premiums does not matter, nor does the difference in time to maturity.

For Further Reference:

(Study Session 14, Module 44.1, LOS 44.a)

CFA® Program Curriculum: Volume 5, page 108

Question #90 of 120

Question ID: 1214519

A synthetic collateralized debt obligation is backed by a portfolio of:

A) **credit default swaps.**
B) structured securities.
C) bonds and other CDOs.

Explanation

Synthetic CDOs have portfolios of credit default swaps as the underlying collateral.

For Further Reference:

(Study Session 14, Module 45.2, LOS 45.i)

CFA® Program Curriculum: Volume 5, page 217

Question #91 of 120

Question ID: 1214516

Which of the following forward rates can be used to construct a forward yield curve?

A) 1-year and 2-year forward rates one year from now.
B) **1-year forward rates one year and two years from now.**
C) 1-year forward rate one year from now and 2-year forward rate two years from now.

Explanation

A forward yield curve is composed of forward rates of the same tenor at different future periods.

For Further Reference:

(Study Session 14, Module 44.4, LOS 44.i)

CFA® Program Curriculum: Volume 5, page 140

Question #92 of 120

Question ID: 1214513

With respect to fixed income markets, the "grey market" refers to trading in:

A) bearer bonds.
B) **bonds that have not yet been issued.**
C) bonds that were issued in private placements.

Explanation

"Grey market" refers to trading of bonds on a when-issued basis.

For Further Reference:

(Study Session 14, Module 43.1, LOS 43.c)

CFA® Program Curriculum: Volume 5, page 66

Question #93 of 120

Question ID: 1214521

Annual Macaulay duration is *least accurately* interpreted as the:

- **A)** weighted average number of years until a bond’s cash flows are scheduled to be paid.
- **B)** **approximate percentage change in a bond’s value for a 1% change in its yield to maturity.**
- **C)** investment horizon at which a bond’s market price risk and reinvestment risk exactly offset.

Explanation

Modified duration is the approximate percentage change in a bond's value for a 1% change in its YTM. Macaulay duration is the weighted average number of periods until a bond's cash flows are scheduled to be paid and represents the investment horizon at which a bond's market price risk and reinvestment risk exactly offset.

For Further Reference:

(Study Session 15, Modules 46.1, 46.3, LOS 46.b, 46.k)

CFA® Program Curriculum: Volume 5, page 248

CFA® Program Curriculum: Volume 5, page 279

Question #94 of 120

Question ID: 1214520

Chris Renburg owns the following portfolio of option-free bonds:

Par value	Full price	Duration
$3,000,000	$2,400,000	4.625
$3,500,000	$3,600,000	7.322
$1,500,000	$1,200,000	9.300
$8,000,000	$7,200,000	

The duration of Renburg's bond portfolio is *closest* to:

- **A)** 6.6.
- **B)** **6.8.**
- **C)** 7.0.

Explanation

Portfolio duration is the weighted average of component securities, using full prices:

(2,400,000 / 7,200,000) × 4.625 + (3,600,000 / 7,200,000) × 7.322 + (1,200,000 / 7,200,000) × 9.3 = 6.753.

For Further Reference:

(Study Session 15, Module 46.2, LOS 46.f)

CFA® Program Curriculum: Volume 5, page 265

Question #95 of 120

Question ID: 1214524

For bonds of the same credit rating, compared to corporate bonds, the default rate for municipal bonds has historically been:

A) **lower.**
B) higher.
C) the same.

Explanation

For bonds with the same credit rating, default rates for municipal bonds have been lower than those of corporate bonds.

For Further Reference:

(Study Session 15, Module 47.2, LOS 47.j)

CFA® Program Curriculum: Volume 5, page 347

Question #96 of 120

Question ID: 1214523

Ann Lloyd, CFA, observes that a 3-year senior unsecured bond of Hawk, Inc. has a rating of Baa3/BBB– and a 3-year senior unsecured bond of Osprey, Inc. has a rating of Ba1/BB+. Based only on this information, Lloyd can *most appropriately* conclude that:

A) credit risk is greater for the Osprey bond than for the Hawk bond.
B) loss severity is greater for the Osprey bond than for the Hawk bond.
C) **the Hawk bond is investment grade and the Osprey bond is non-investment grade.**

Explanation

The classifications "investment grade" and "non-investment grade" are based on ratings from recognized credit rating agencies. Bonds rated Baa3/BBB– or higher are classified as investment grade, while bonds rated Ba1/BB+ or lower are classified as non-investment grade. However, an analyst should not rely exclusively on credit ratings to draw conclusions about the credit risk or loss severity of bond investments.

For Further Reference:

(Study Session 15, Module 47.1, LOS 47.e)

CFA® Program Curriculum: Volume 5, page 315

Question #97 of 120

Question ID: 1214512

Contingent convertible bonds are described *most accurately* as those which, if a specified event occurs:

A) become convertible to equity.
B) **convert automatically to equity.**

C) increase the equity conversion ratio.

Explanation

Contingent convertible bonds are converted automatically to common stock if a specified event occurs.

For Further Reference:

(Study Session 14, Module 42.2, LOS 42.f)

CFA® Program Curriculum: Volume 5, page 37

Question #98 of 120

Question ID: 1214517

If a callable bond has an option-adjusted spread (OAS) of 75 basis points, this *most likely* suggests:

A) **the bond has a zero-volatility spread greater than 75 basis points.**

B) the implied cost of the call option is the bond's nominal spread minus 75 basis points.

C) the 75 basis points represent the investor's compensation for credit risk, liquidity risk, and volatility risk.

Explanation

For a bond with an embedded call option, the OAS is less than its zero-volatility spread by the option cost. Therefore, the zero-volatility spread is greater than the OAS for callable bonds. If the embedded call option has any value to the issuer, a callable bond with an OAS of 75 basis points will have a *Z*-spread that is greater than 75 basis points.

Because the OAS represents the bond's spread to the spot yield curve excluding the effect of the embedded option, it does not include any compensation for the volatility risk related to the option. The implied cost of an embedded option is the difference between the bond's zero-volatility spread (not the nominal spread) and its OAS.

For Further Reference:

(Study Session 14, Module 44.5, LOS 44.k)

CFA® Program Curriculum: Volume 5, page 147

Question #99 of 120

Question ID: 1214522

Wendy Jones, CFA, is reviewing a current bond holding. The bond's duration is 10 and its convexity is 200. Jones believes that interest rates will decrease by 100 basis points. If Jones's forecast is accurate, the bond's price will change by approximately:

A) –8.0%.

B) +8.0%.

C) **+11.0%.**

Explanation

You can answer this question without calculations. A decrease in interest rates must cause the price to increase. Because duration alone will underestimate a price increase, the price must increase by more than 10%.

$$\text{percentage change in price}$$
$$=[-\text{duration} \times \Delta\text{YTM}]+\frac{1}{2}\left[\text{convexity} \times (\Delta\text{YTM})^2\right] \times 100$$
$$= [(-10)(-0.01)] + \frac{1}{2}\left[(200)(-0.01)^2\right] = 0.11 = 11\%$$

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.i)

CFA® Program Curriculum: Volume 5, page 270

Question #100 of 120

Question ID: 1214531

Questions 100 through 106 relate to Derivatives. (10.5 minutes)

In a one-period binomial model for option pricing, the probabilities of an up-move or a down-move are:

A) equal.
B) assumed by the modeler.
C) calculated from the model inputs.

Explanation

Risk-neutral pseudo-probabilities are calculated using the risk-free rate and the sizes of an up-move and down-move of the underlying asset.

For Further Reference:

(Study Session 16, Module 49.4, LOS 49.n)

CFA® Program Curriculum, Volume 5, page 496

Question #101 of 120

Question ID: 1214525

Roland Carlson owns a portfolio of large capitalization stocks. Carlson has a positive long-term outlook for the stock market, but would like to protect his portfolio from any sudden declines in the stock market, without selling his holdings. The *most likely* way for Carlson to achieve his objective of limiting the downside risk of his portfolio is to:

A) sell put options on the S&P 500.
B) sell an S&P 500 futures contract.
C) buy an S&P 500 forward contract.

Explanation

Losses on Carlson's portfolio of large cap stocks can be offset by gains on a short position in a futures contract. (Gains on the portfolio would be offset by futures losses.) He could also *buy* put options on the S&P 500. A long position in an S&P 500 forward contract would not offer any downside protection.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.c)

CFA® Program Curriculum, Volume 5, page 394

Question #102 of 120

Question ID: 1214529

The time value of a put option on an asset that provides no cash flows would *most likely* be increased by:

A) an increase in the exercise price.

B) an increase in the asset's price volatility.

C) a decrease in the value of the underlying asset.

Explanation

An increase in volatility will increase the value of a put option but will not change its intrinsic value, so it is the time value that increases. Changes in the exercise price or the value of the underlying asset will change an option's intrinsic or exercise value.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.k)

CFA® Program Curriculum, Volume 5, page 482

Question #103 of 120

Question ID: 1214528

At time *t*, prior to its settlement date at time *T*, the value V_t of a long forward with a price of *F* will be related to the spot price, *S*, of an asset that has a zero net cost of carry by:

A) $V_t = F - S/(1 + Rf)^{(T-t)}$

B) $V_t = (S - F)/(1 + Rf)^{(T-t)}$

C) $\mathbf{V_t = S - F/(1 + Rf)^{(T-t)}}$

Explanation

The value of a long position in a forward contract prior to settlement (expiration) is:

$V_t = S - F/(1 + Rf)^{(T-t)}$ when the net cost of carry is zero.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.d)

CFA® Program Curriculum, Volume 5, page 469

Question #104 of 120

Question ID: 1214527

A put option with an exercise price of $75 sells for a premium of $10. At expiration, the put buyer may experience a loss:

A) of as much as $10.

B) of as much as $65.

C) that is theoretically unlimited.

Explanation

The greatest loss a put buyer can experience is the premium paid. If the price of the underlying asset decreased to zero, the put *writer* would experience a loss of ($0 – $75) + $10 = –$65.

For Further Reference:

(Study Session 16, Module 48.2, LOS 48.d)

CFA® Program Curriculum, Volume 5, page 406

Question #105 of 120

Question ID: 1214526

A financial instrument with a payoff that depends on a specified event occurring is *most accurately* described as:

A) an option.
B) a default swap.
C) a contingent claim.

Explanation

Contingent claims are contracts with payoffs that depend on a specified event occurring. Options and credit default swaps are examples of contingent claims, but neither of these terms describes all contingent claims.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.b)

CFA® Program Curriculum, Volume 5, page 387

Question #106 of 120

Question ID: 1214530

An investor uses options on a stock to create a synthetic short position in a risk-free bond that will pay the exercise price at option expiration. To create this position, the investor will buy:

A) a put option.
B) a call option.
C) the underlying stock.

Explanation

Using the put-call parity relationship, a synthetic short position in a risk-free bond that pays the exercise price at expiration can be created by buying a call, writing a put, and taking a short position in the stock.

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.l)

CFA® Program Curriculum, Volume 5, page 491

Question #107 of 120

Question ID: 1214532

Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)

A leveraged buyout fund is evaluating Siena Company relative to its peer companies. Siena is *most likely* a good candidate for a management buy-in if it has:

A) **higher cash flow and less capable managers than its peers.**
B) lower cash flow and more capable managers than its peers.
C) higher cash flow and more capable managers than its peers.

Explanation

In a management buy-in, a leveraged buyout (LBO) fund replaces the existing managers of a portfolio company with a new team it believes can increase the value of the company. Companies with high cash flow are attractive candidates for LBOs because their cash flow can help service the debt issued to finance the LBO. A company with high cash flow and capable managers is a potential candidate for a management buyout (MBO), a transaction in which the managers participate and stay on after the company goes private.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #108 of 120

Question ID: 1214533

The value of an existing single-family home used for residential purposes will *most likely* be calculated using the:

A) cost approach.
B) income approach.
C) **sales comparison approach.**

Explanation

An existing single-family home for residential purposes will most likely be valued using the sales comparison method.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 10

Question #109 of 120

Question ID: 1214534

An investment in a hedge fund with a 2-and-20 fee structure has increased in value each period and earned a return of 8% net of management fees in 20x7. Under which of the following provisions would incentive fees for 20x7 be the highest?

A) 5% hard hurdle rate and a high water mark provision.
B) **6% soft hurdle rate and a high water mark provision.**
C) 7% hard hurdle rate and no high water mark provision.

Explanation

With a soft hurdle rate, the incentive fee is a percentage of the entire return once the hurdle rate is met. With a hard hurdle rate, the incentive fee is a percentage of return in excess of the hurdle rate. A high water mark does not affect the incentive fee for an account that has increased in value each period. A soft hurdle rate would result in incentive fees of 20% × 8% = 1.6% of assets; a 5% hard hurdle rate would result in incentive fees of 20% × (8% − 5%) = 0.6% of assets; and a 7% hard hurdle rate would result in incentive fees of 20% × (8% − 7%) = 0.2% of assets.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.d)

CFA® Program Curriculum, Volume 6, page 20

Question #110 of 120

Question ID: 1214535

Because of survivorship bias, hedge fund data are *most likely* to:

- **A)** overstate returns and overstate risk.
- **B)** **overstate returns and understate risk.**
- **C)** understate returns and overstate risk.

Explanation

Survivorship bias in hedge fund returns contributes to overstatement of performance and understatement of risk.

For Further Reference:

(Study Session 17, Modules 50.1, 50.2, LOS 50.c, 50.f)

CFA® Program Curriculum, Volume 6, page 13

CFA® Program Curriculum, Volume 6, page 57

Question #111 of 120

Question ID: 1214536

A commodity market is in contango if the spot price is:

- **A)** higher than futures prices.
- **B)** equal to futures prices.
- **C)** **lower than futures prices.**

Explanation

A commodity market is contango if the futures price is higher than the spot price.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #112 of 120

Question ID: 1214537

Infrastructure investments as an asset class:

A) are illiquid.

B) **include social infrastructure.**

C) refer to assets operated by the entity constructing the asset.

Explanation

Social infrastructure investments refers, for example, to schools, prisons, and healthcare facilities. While some infrastructure investments are illiquid, relatively liquid investment vehicles have developed for investors seeking exposure to the asset class.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #113 of 120

Question ID: 1214538

A hedge fund that employs a fundamental growth strategy using equity securities is *most likely* to seek out shares of companies that are:

A) undervalued only.

B) either undervalued or overvalued.

C) **growing revenues and earnings rapidly.**

Explanation

Fundamental growth refers to investing in companies that are experiencing high growth and for which the fund managers anticipate significant capital appreciation.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #114 of 120

Question ID: 1214545

Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)

Which of the following fintech applications is *most likely* to use distributed ledger technology?

A) **Real-time clearing and settlement of securities trades.**

B) High-frequency trading to take advantage of intraday price discrepancies.

C) Automated review of employees' voice communications to detect misconduct.

Explanation

Real-time clearing and settlement is a potential application of distributed ledger technology because it would require an identical record of transactions to be visible to multiple parties. Neither algorithmic trading nor natural language processing necessarily involves distributed ledger technology.

For Further Reference:

(Study Session 19, Module 57.1, LOS 57.c, 57.d)

CFA® Program Curriculum, Volume 6, page 411

CFA® Program Curriculum, Volume 6, page 416

Question #115 of 120

Question ID: 1214540

Compared to a normal distribution, historical returns on major asset classes in developed markets have exhibited:

A) less frequent large positive deviations.
B) **more frequent large negative deviations.**
C) the expected frequency of large deviations.

Explanation

Historical data from global asset markets show that returns distributions exhibit negative skewness and positive excess kurtosis. Large negative deviations, and large deviations in general, have been more frequent than would be expected if returns were normally distributed.

For Further Reference:

(Study Session 18, Module 52.1, LOS 52.c)

CFA® Program Curriculum, Volume 6, page 133

Question #116 of 120

Question ID: 1214539

Which of the following pooled investments is likely to require the smallest minimum investment amount?

A) Wrap fee account.
B) Market neutral fund.
C) **Closed-end mutual fund.**

Explanation

Closed-end mutual fund shares can be purchased for market-determined prices. Hedge funds (which include market neutral funds) and wrap fee (separately managed) accounts typically require substantial minimum investment amounts.

For Further Reference:

(Study Session 18, Module 51.2, LOS 51.f)

CFA® Program Curriculum, Volume 6, page 104

Question #117 of 120

Question ID: 1214542

All portfolios that lie on the capital market line:

A) **contain the same mix of risky assets unless only the risk-free asset is held.**

B) have some unsystematic risk unless only the risk-free asset is held.

C) contain at least some positive allocation to the risk-free asset.

Explanation

All portfolios on the CML include the same tangency portfolio of risky assets, except the intercept (all invested in risk-free asset). The tangency portfolio contains none of the risk-free asset and "borrowing portfolios" can be constructed with a negative allocation to the risk-free asset. Portfolios on the CML are efficient (well-diversified) and have no unsystematic risk.

For Further Reference:

(Study Session 18, Module 53.1, LOS 53.b, 53.c)

CFA® Program Curriculum, Volume 6, page 192

CFA® Program Curriculum, Volume 6, page 204

Question #118 of 120

Question ID: 1214544

From a high of $180, a stock price decreases to a low of $100 and then begins increasing. A technical analyst states that she expects resistance levels to emerge at $140, $150, and $153.33. This analyst is *most likely* forecasting these resistance levels based on:

A) **Fibonacci numbers.**

B) an inverse head and shoulders pattern.

C) moving average convergence/divergence lines.

Explanation

The forecast resistance levels are one-half, five-eighths, and two-thirds of the price decrease from $180 to $100. All of these are Fibonacci ratios. Projecting from the breakout of an inverse head and shoulders pattern would more likely suggest a single price target or range than three different specific targets. Moving average convergence/divergence lines are unlikely to be used for price targeting because they are not on the same scale as prices.

For Further Reference:

(Study Session 19, Module 56.1, LOS 56.d, 56.e, 56.g)

CFA® Program Curriculum, Volume 6, page 355

CFA® Program Curriculum, Volume 6, page 366

CFA® Program Curriculum, Volume 6, page 384

Question #119 of 120

Question ID: 1214543

Stephanie Dell is evaluating two stocks (X and Y) using the capital asset pricing model. Dell predicts that the betas for the two stocks will be identical but that the unsystematic risk for Stock X will be much higher than for Stock Y. According to the capital asset pricing model, in equilibrium:

A) Stock X will have a higher expected return than Stock Y but a standard deviation equal to Stock Y.

B) **Stock X will have a higher standard deviation than Stock Y but an expected return equal to Stock Y.**

C) both the expected return and standard deviation for Stock X will be higher than Stock Y.

Explanation

The equation for the capital asset pricing model is:

$$E(R_i) = R_F + \beta_i[E(R_m) - R_F]$$

Beta measures the sensitivity of the stock's returns to changes in the returns on the market portfolio and is a standardized measure of the stock's systematic or non-diversifiable risk. As indicated by the CAPM equation, the expected return for any stock is related to its beta. In contrast, unsystematic risk does not affect the CAPM expected return. Therefore, according to the CAPM, expected returns are identical for assets with identical betas. Stock X has identical systematic risk but greater unsystematic risk than Stock Y, resulting in greater total risk (standard deviation).

For Further Reference:

(Study Session 18, Module 53.2, LOS 53.f)

CFA® Program Curriculum, Volume 6, page 213

Question #120 of 120

Question ID: 1214541

Boswell is less risk-averse than Johnson. Using the same capital allocation line for Boswell and Johnson, Boswell will have:

A) a higher risk aversion coefficient than Johnson.

B) steeper risk-return indifference curves than Johnson.

C) **an optimal portfolio with a higher expected return than Johnson.**

Explanation

Compared to a more risk-averse investor, a less risk-averse investor will have a lower risk aversion coefficient, flatter risk-return indifference curves, and an optimal portfolio with greater risk and a higher expected return.

For Further Reference:

(Study Session 18, Module 52.3, LOS 52.i)

CFA® Program Curriculum, Volume 6, page 166